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Medical ChatBot Assistance for Primary Clinical Guidance using Machine Learning Techniques

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Abstract – Hospitals serve as critical hubs for global healthcare, offering essential medical services. However, a pressing issue is the lack of 24/7 medical facilities in rural areas, particularly during the overnight hours. To bridge this gap, we propose the Robotic Medical Support ChatBot (RMSCB) system. This innovative online platform leverages machine learning to autonomously predict medical diagnoses and provide temporary solutions, accessible at any time. The RMSCB system comprises three key stages: pre-processing, training, and classification. In the pre-processing stage, relevant keywords are extracted using the Pre-Fixed Stopping Words Model (PFSWM). The training stage categorizes pre-processed questions into classes based on the Pre-Fixed Class Label Model (PFCLM), associating medical solutions via the Pre-Fixed Answer and Question Model (PFAQM). In the classification stage, the RMSCB processes clinical questions from patients via mobile or personal systems, mapping and identifying solutions based on probability. Experimental results demonstrate the efficacy of the RMSCB system, offering timely first aid medication information and alleviating the need for immediate physician consultation. This technological solution addresses critical healthcare gaps, especially in regions with limited access to continuous medical services.

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1. Introduction

In our contemporary society, healthcare assumes a pivotal role amidst the hustle and bustle of daily life. The majority of individuals find themselves deeply immersed in various commitments at home and work, often further distracted by the pervasive influence of the Internet, including platforms like YouTube, OTTP, Facebook, WhatsApp, Twitter, and E-News. Regrettably, amidst these busy lives, health often takes a backseat, with a notable

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reluctance among individuals to seek medical attention at nearby hospitals. This avoidance can be attributed to a confluence of factors, such as the considerable distance from health centers, infrequent availability of physicians, skepticism regarding medical policies, and heightened concerns about the financial implications of treatment and medication.Over the past decade, the Information Technology (IT) sector has undergone transformative growth, impacting a multitude of industries. This technological revolution has permeated diverse sectors, from automobile and tourism to banking, online commerce, aviation, space research, geography, rocket science, weather forecasting, net banking, medicine, defense, education, networking, automation systems (including home appliances), ChatBot technology, and security systems, among others. In this context, ChatBot [1][2] systems have emerged as versatile and efficient real-time online customer helplines, providing textual interactions across various domains, including education, banking, medicine, and commerce [3][4][5].

Despite these advancements, traditional ChatBot systems, especially those in the healthcare sector, face limitations when it comes to effectively addressing patient queries related to various illnesses. This gap becomes particularly pronounced during non-standard hours when immediate medical attention may be crucial [6][7][8]. To address these limitations and enhance healthcare accessibility, this paper introduces the Robotic Medical Support ChatBot (RMSCB) system [9].The RMSCB is designed to empower patients by offering autonomous predictions of medical diagnoses and providing short-term solutions without the need for direct physician intervention. Importantly, this service is available 24 hours a day and is offered free of charge. The system is meticulously crafted to overcome the shortcomings of existing medical ChatBot systems, ensuring that individuals receive prompt and reliable first aid medication information even during unconventional hours. This innovative approach seeks to redefine healthcare assistance, catering to the diverse needs of individuals in a comprehensive and technologically advanced manner.

2. Related Work

Several researchers have explored the use of artificial intelligence-based chatbots to provide medical advice and improve healthcare access. Article [10] focused their research on an artificial intelligence-based medical chatbot. This study addresses the critical need for accurate medical advice, considering that doctors may occasionally make errors in symptom observation, whereas a specialized machine is inherently less prone to such mistakes. In [11], highlighted the emergence of chatbots as a promising platform for healthcare services, evident in the increasing availability of publicly accessible chatbots involved in prevention, diagnosis, and treatment services. This system scrutinizes how chatbots approach design aspects crucial to healthcare service provision, emphasizing Human-AI interaction and transparency in AI automation and decision-making.In [12], presented an innovative chatbot application leveraging machine learning for medical diagnosis. This chatbot aims to diagnose diseases and provide probable outcomes before consulting a doctor, thereby reducing consultation costs and enhancing patient medical knowledge. The chatbot utilizes pattern matching to predict patient symptoms. Additionally, the application offers an SOS button and a map displaying nearby doctors, ensuring immediate assistance for the users.

In[13]have designed a type of clinical related chatbot system for various patients to reduce the healthcare costs and improve accessibility of medical knowledge. Certain chatbots acts as a medical reference books, whichhelpsthepatientknowmoreabouttheirdiseaseandhelpstoimprovetheir health [14,15,16].Theusercan achieve the real benefit of a chatbot only when it can diagnose all kind of disease andprovidenecessary information. In [17]have implemented online based medication first aid system that used to the patents to live a good life health care is vital factor for individuals. But it is not so easy to consult the doctor in case of any health-related issues. The proposed idea is to create chatbot for healthcare as virtual assistant using Artificial Intelligence. Artificial intelligence integrated healthcare chatbot which will diagnose and provides the common health issues details before consulting a doctor. In [18] have designed with increasing population India, increasing birth rate and decreasing death rate due to advancement in the medical field it's found that numbers of doctors are less to serve the need of the increasing population. This scenario can be better understood while walking through the city's government hospitals where the less availability of the doctors is the major cause behind the improper treatment of the patients and in certain scenario the resultant death. Sometime even doctors can make mistake in providing the correct treatment result in

death of patient [19, 20].

3. Proposed System

This section delves into the operational intricacies of the proposed Robotic Medical Support ChatBot (RMSCB) system, with the primary objective of encouraging individuals with various health concerns to autonomously predict the medical analysis of their ailments without relying on a physician, leveraging machine learning techniques. The RMSCB system is structured into four distinct stages: pre-processing, training, and classification. In the pre-processing stage, the system extracts pertinent keywords from each medical question within the extensive set of questions presented in unstructured text format. This extraction is achieved through the application of the Pre-Fixed Stopping Words Model (PFSWM). Subsequently, building upon the outcomes of the pre-processing stage, the system identifies a finite number of classes within the pre-processed question set using a probability-based approach and the Pre-Fixed Class Label Model (PFCLM). Following this, the RMSCB associates specific medical solutions or answers, in descriptive form, with their respective class labels, employing the Pre-Fixed Answer and Question Model (PFAQM).

Moving to the classification stage, the RMSCB pre-processes medical questions from the patient side and maps these pre-processed questions onto a trained dataset equipped with class labels. Finally, through probability-based classification, the system discerns the particular class and provides the corresponding diagnosis solution for the patient's ailment. The functionality of each model within the proposed RMSCB system is visually represented in Figure 1, offering a comprehensive overview of the intricate processes involved in enabling individuals to independently analyze their health conditions through this innovative technological solution.

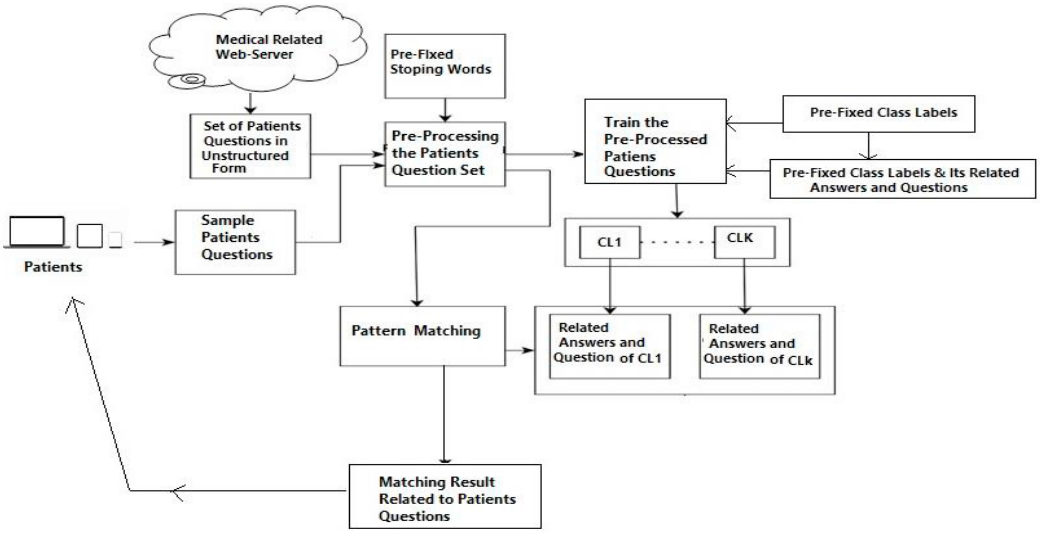


Figure 1: Proposed RMSCB System Architecture

3.1. Medical Question Pre-Processing Stage

In this sub section, the RMSCB system uses to limit the number of keywords over the medical related raw question set with lot of different illness questions with various keywords inform of text via to trace and filter the unrelated words including verb, articles, proposition, and connection words through Pre-Fixed Stopping Words Model (PFSWM).It contains two steps, initially, it accumulates the higher numbers of conversation medical related information's with set of questions (Q) between patients and physician from medical related web server and is defined as $Q = Q_i$, $Q_i = Q_{ij}$ for $i = 1,2,..,r$, and $j = 1,2,..,m$ where Q is the unstructured historical medical

related question set with various questions, r denotes the number of medical related questions in $Q$and$Q_i$denotes the i[th] question in the input question set, , m is described number words in the ith question in Q. Subsequently, it recognizes the stopping words over each individual question in the actual question set$Q$via Pre-Fixed Stopping Words Model (PFSWM)and based on word matching technique then remove the identified stopping words in the respective questions in the question set and it has defined as $\bar{Q} = \bar{Q}_i$for $i = 1,2,..,r$, where $\bar{Q}_i$represents the ith pre-processed medical question in the pre-processed actual question set $\bar{Q}$ with $r$questions. The pre-Fixed Stopping Word Model (PFSWM) is containing restricted number of commonly used simple English words including verbs, articles, proposition and conjunction respectively.

3.2. Question Set Training Stage

This subsection, the RMSCB system is separate the content of previous stage result with pre-processed question set $\bar{Q}$ into finite number of dissimilar disease related class labels via probability technique based on Pre-Fixed Class Label Model (PFCLM) and Pre-Fixed Answer & Question Model (PFAQM) related to class label of the disease. The PFCLM contains finite number of various human illnesses as a class labels including Fever, Cold, Cough, Headache etc., and it has defined asD= D_f,for $f = 1,2,..,l$, where, D is denoted the finite number of human disease label set,D_f is the fth disease label in the D and l denotes the number of labels in the D. Similarly, the PFAQM is containing action or medical first aid information for various illness related to pre-fixed various disease names in PFCLM and it has defined as S= S_f,for $f = 1,2,..,l$, where, S is denoted finite number of solutions or answers to respective disease labels in D,S_f is the fth solution or medication information to f[th] disease label in the D and$l$ denotes the number of Solutions in the S. The training stage is iteratively computes probability between each individual medical question in $\bar{Q}$ and over all the medical labels in the$D$ then find the respective class label to the respective question via pick the f[th] label $D_f$in $D$with maximum probability to i[th] question $\bar{Q}_i$in $\bar{Q}$and it has defied in the Equation (1) as:

$$DL(\bar{Q}_i) = \underset{f=1,2,..l}{Max}\{P(\bar{Q}_i \mid D_f) \,\big|\, \forall D_f \in D, \forall \bar{Q}_i \in Q\} \tag{1}$$

$$P(\bar{Q}_i \mid D_f) = \frac{Count\,of\ D_f\ is\ occured\ in\ \bar{Q}_i}{\left|\bar{Q}_i\right|} \tag{2}$$

Where, $\left|\bar{Q}_i\right|$ is denoted the number of words in the i[th] medical question in the pre-processed question set $\bar{Q}_i = \bar{Q}_{ij}$ for $j = 1,2,..,m$. Finally, repeat the above process until identifies the disease label to all the pre-processed question in the question set $\bar{Q} = \bar{Q}_i$for $i = 1,2,..,r$ and obtained the trained medical information including question, related class labels and diagnosis solutions related to the class label respectively as defined in set $\overline{TQ} = \overline{TQ}_i$ for $i = 1,2,..,r$, where $\overline{TQ}$ is the trained medical question set which is containing medical question and its related class label with related medical diagnosis information.

3.3. Pattern Matching Stage

In thissub chapter, the proposed RMSCB system identifies the test query or question related to the medication diagnosis of the specific disease without class label that belongs into which disease class in the trained question set $\overline{TQ}$ viaprobability based classification [21].Initially, it pre-processes the test medical question (tq)related to the disease without specific class label throughPre-Fixed Stopping Words Model (PFSWM) which is already conversed in the chapter (3.1). Next, it computes the counts of words over the content of the existing various disease class labels $(DL = DL_e)$ in the training medical question set $\overline{TQ} = \overline{TQ}_i$ for $i = 1,2,..,r$. In the final step, it calculates the conditional probability $P(DL_e|\bar{tq})$ of pre-processed test question $\bar{tq}$over the content of the disease label in the training question set $\overline{TQ}$, where DL_e is the eth disease label in the training question set $DL_e \subseteq \overline{TQ}$ and is defined in the Equation (3) as:

$$P(DL_e|\bar{tq}) = P(DL_e)\prod_{j=1}^{m} P(\bar{tq}_j|DL_e)\ |\forall DL_e \in DL, \forall \bar{tq}_j \in \bar{tq} \quad (3)$$

Where, $P(DL_e)$ is the prior probability of e[th] disease label in the trained question set $\overline{TQ}$ and is defined inthe Equation (4) as:

$$P(DL_e) = \left\{ \frac{W}{|\overline{TQ}|} \,|\, where\ 0 > W < r \right\} \quad (4)$$

Here, W denotes the number of trained medical questions which are belonging in the e[th] disease class label DL_e in the training question set $\overline{TQ}$, $|\overline{TQ}|$ represents the number of medical questions in the training question set $\forall \overline{TQ}_i \in \overline{TQ}$ for $i = 1,2,..,r$, $\overline{tq_j}$ denotes the j[th] text word in the pre-processed test medical question $\overline{tq} = \overline{tq}$ for $j = 1,2,..,m$. Next, it identifies the test medical question $\overline{tq}$ is matched with which disease class DL_e in the trained question set $\overline{TQ}$ via pick the class label which is obtained maximum conditional probability with test question$P(DL_e|\bar{tq})$and it is defined in the equation (5).

$$DL(\overline{tq}) = \{Max_{e=1,2,..k}\left\{P(DL_e \mid \overline{tq}\right\}, |\forall DL_e \in DL\} \quad (5)$$

In the final step, it follows the pattern matching result DL_e and subsequently it generates the respective medical diagnosis information or answer DA_e to the respective medical test question (tq) and is defined as $DA = DA_e$ for $e = 1,2,..,k$, where DA_e is the e[th] diagnosis answer or information in the diagnosis answer set DA with k different answers.

ALGORITHM: PBTC

Input: Medical Question Set $Q = \{Q_1, Q_2, .., Q_r\}$, Test Question tq, PFSWM, PFCLM,PFAQM

Output: Trained Medical Question Set $\overline{TQ} = \overline{TQ}_i$, Pattern Matching Result DL_e and DA_e

Begin

1. Pre-process medical question set Q= Q_i for $i = 1,2,..,r$ via PFSWM and obtained in $\bar{Q} = \bar{Q}_i$ for $i = 1,2,..,r$
2. Separate the pre-processed question set $\bar{Q} = \bar{Q}_i$ into k disease class label via PFCLM and PFAQM
3. Obtained training result in $\overline{TQ} = \overline{TQ}_i$ for $i = 1,2,..,k$
4. Compute counts of number of words over the each individual class in training question set $DL = DL_e$ for $e = 1,2,..,k$ for using Equations (1 & 2).
5. Calculates probability $P(DL_e)$ of each individual class label $DL = DL_e$ in trained question set $\overline{TQ} = \overline{TQ}_i$using equation (4)
6. Computes conditional probability$P(DL_e|\bar{tq})$ between pre-processed test question $\bar{tq}$ and content of the k classes in the trained question set $\overline{TQ} = \overline{TQ}_i$ and $DL = DL_e$ using Equation (3)
7. Recognizes the test question $\bar{tq}$is matched to which class label among k class labels in the trained question set $\overline{TQ} = \overline{TQ}_i$ using equation (5)
8. Generates respective diagnosis information DA_e related to the matched class label DL_e and its test question (tq).

End

4. Result and Discussion

This section presents the findings from the investigative evaluation of the proposed Robotic Medical Support ChatBot (RMSCB) system, focusing on its performance in handling medical diagnosis-related healthcare questions and answers. To assess its efficacy, a substantial amount of medical-related conversational data was gathered, encompassing interactions between diverse patients and physicians sourced from medical-related web servers. A subset of this dataset, including a minimal number of questions, is outlined in Table 1. The details of medical diagnosis-related questions are highlighted in Table 1, providing insights into the system's responsiveness and effectiveness in addressing healthcare inquiries. $Q = Q_i$ for $i = 1,2,..,$ and mentioned number of words in the respective questions $|Q_i|$.

Table 1 Actual Medical Diagnosis Related Question Set

Medical Question ID	Medical Question Q_i	Question Size $\|Q_i\|$
1	I am having high fever, what is the medicine	9
2	I am having fever, how to reduce the fever and what is diagnosis	13
3	I am having cold problem, what is the remedies to reduce the problem	4
4	I am having high cold, what is the medication?	4
5	I am having cough, how to reduce the cough and what is diagnosis	8
6	I am having cold problem, what is the remedies to reduce the problem	4
7	I am having high cold, what is the medication?	4
8	I am having headache, what is the better medicine	9

In the beginning stage, the RMSCBsystem haspre processedtheinput medical question set Q with$r$questionsrelated to medical diagnosis which are shown in the Table 1 throughpre fixed stopping words model (PTPM-1) has presented in the Table 2. The pre processed result of question set $\bar{Q} = \bar{Q}_i$for $i = 1,2,..,$ has illustrated in the Table 3.

Table 2 Pre-Fixed Stopping Word Model (PFSWM)

PFSWM
is, the, with, are, there, has, had, their, it, and, but, they, three, this, that, then, will, should, could, can, did etc.

Table 3 Outcome of Pre Process Operation of Input Question Set (Q)

Medical Question ID	Medical Question $\overline{Q}_i$	Question Size $\|\overline{Q}_i\|$
1	I have high fever medicine	5
2	I have fever reduce fever diagnosis	6
3	I have cold remedies	4
4	I high cold medication	4
5	I have cough diagnosis	4
6	I have cold remedies	4
7	I have high cold medication	5
8	I have high headache medicine	5

Next, the RMSCB system divides the preprocessed medical question set $\overline{Q}_i$ into k finite number of medical classes with various medical disease names as a labels to the classes including fever, cold, cough and headache respectively $(DL = DL_e)$ and also enclosed the respective medical diagnosis solution to corresponding class label $(DA = DA_e)$ in the training question set $\overline{TQ}$based on probability technique has discussed in the sub chapter (3.2) through PFCLM and PFAQM as presented in the Table 4. The outcome of the training question set $\overline{TQ}$ with medical questions and its respective medical class labels with its respective diagnosis solutions as presented in the Table 5.

Table 4 Pre Fixed Class Label Model (PFCLM)& Pre Fixed Answer and Question Model (PFAQM)

Pre Fixed Class Label Model (DL_e)	**Pre Fixed Answer and Question Model** (DA_e)
Fever	First You have to Confirm Your Body Temperature& Follow the Medicine First Aide Information: If the measured body temperature is falling between $98^0 C$ to $101^0 C$ (degree) then to take any one medicine as given and then must meet the Physician without fail: For Adult - "Paracetamol 500mg /Nimesulide 100mg /Amoxycillin 250mg" For Child –Tylenol, Advil, Mortin
Cold	If you are having high cold problem then you have to take any one below listed medicine: For Adult – Robitussin, Mucinex DM For Child – Dextr, Zarbees
Cough	If you are having high cold problem then you have to take any one below listed medicine: For Adult – Robitussin, Mucinex DM For Child – Dextr, Zarbees
Headache	If you are having high Headache problem then you have to take any one below listed medicine: For Adult – For Child - the below

Table 5 Outcome of Training Process $\overline{TQ}$ Over the Pre-Processed Question Set $\overline{Q}$

Medical Question ID	**Trained Medical Question Set** $\overline{TQ}_i$	**Diagnosis Class Label** (DL_e)	**Medical Diagnosis Solution or Answer** (DA_e)
1	I have high fever medicine	Fever	First You have to Confirm Your Body Temperature& Follow the Medicine First Aide Information: If the measured body temperature is falling between $98^0 C$ to $101^0 C$ (degree) then to take any one medicine as given and then must meet the Physician without fail: For Adult - "Paracetamol 500mg /Nimesulide 100mg /Amoxycillin 250mg" For Child –Tylenol, Advil, Mortin
2	I have fever fever diagnosis	Fever	
3	I have cold remedies	Cold	If you are having high cold problem then you have to take any one below listed medicine: For Adult – Robitussin, Mucinex DM For Child –Dextr, Zarbees
4	I have high cold medication	Cold	
5	I have cough cough diagnosis	Cough	If you are having high Cough problem then you have to take any one below listed medicine: For Adult – Robitussin, Mucinex DM For Child – Dextr, Zarbees
6	I have cold remedies	Cold	If you are having high cold problem then you have to take any one below listed medicine: For Adult – Robitussin, Mucinex DM For Child – Dextr, Zarbees
7	I have high cold medication	Cold	
8	I have high headache medicine	Headache	If you are having high Headache problem then you have to take any one below listed medicine: For Adult – For Child - the below

Table 6 Test Medical Question (tq)

Test Question ID	**Actual Test Question** (tq)	**Pre-processed Test Question** $(\overline{tq})$
tq_1	I am having high fever	I high fever

Finally, the RMSCB system has tested sample test question (tq) without class label over the training question set $(\overline{TQ})$ and identified the class label with diagnosis answer to the respective test medical question as presented in the Table 6 and Table 7. Investigation outcomes revealed in the Table 6 and Table 7 that the RMSCB system is fit to sick people that to avoid the complicated situation and to manage sick people treatments via first aid medication information without meet the physician in their health carehub.

Table 7 Pattern Matching Result for Test Question ($\overline{tq}$)

Pre processed Test Question ($\bar{t}q$)	Prior Probability of $P(Fever$	Prior Probability of $P(Cold)$	Prior Probability of $P(Cough)$	Conditional Probability of $P(fever\|\overline{tq})$	Conditional Probability of $P(cold\|\overline{tq})$	Conditional Probability of $P(cold\|\overline{tq})$	Pattern Matching Result (DL) with Diagnosis solution (DA_e)
I have high fever	0.25	0.5	0.5	0.0029	0.0	0.0	**Fever** / First You have to Confirm Your Body Temperature& Follow the Medicine First Aide Information: If the measured body temperature is falling between 98^0C to 101^0C (degree) then to take any one medicine as given and then must meet the Physician without fail: For Adult - "Paracetamol 500mg /Nimesulide 100mg /Amoxycillin 250mg" For Child -

5. Conclusion

This paper introduces an online medication system called the Robotic Medical Support ChatBot (RMSCB) system, designed to enable individuals with various health concerns to autonomously predict the medical diagnosis of their ailments and receive temporary solutions without the need for a doctor, utilizing machine learning techniques. The RMSCB system comprises four distinct stages: pre-processing, training, and pattern matching. In the pre-processing stage, the system extracts relevant keywords from each health-related question in the chat question set, utilizing the Pre-Fixed Stopping Words Model (PFSWM) to handle unstructured text keywords. Subsequently, building on the outcomes of the pre-processing stage, the system identifies a finite number of classes within the pre-processed question set through a probability-based approach and the Pre-Fixed Class Label Model (PFCLM). Following this, the RMSCB associates specific medical solutions or answers, expressed descriptively, with their respective disease labels, using the Pre-Fixed Answer and Question Model (PFAQM).In the final model, the RMSCB tests medical questions from the patient side, mapping pre-processed test questions onto a trained dataset with class labels. Preliminary results indicate that the RMSCB system is highly suitable for individuals seeking to manage their health proactively, providing first aid prescription information without the need for immediate physician consultation, especially in critical situations.

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